SUMS OF POWERS OF INTEGERS: A COMPLETE FRAMEWORK FOR CLOSED FORMULAS

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Abstract.

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1. Definitions and notations

Allow us to define the mathematical objects we utilize throughout the manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [1], because it is simple and beautiful.

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\begin{aligned}\Sigma^0 n^m &= n^m,\\ \Sigma^1 n^m &= \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,\\ \Sigma^{r+1} n^m &= \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.\end{aligned}$$

We utilize the following notation for falling factorials

Definition 1.2 (Falling factorials). *For integers x, n*

$$(x)_n = \begin{cases} 1, & \text{if } k = 0,\\ x(x-1)(x-2)\cdots(x-n+1) = \prod_{k=0}^{n-1}(x-k), & \text{if } k > 0,\\ 0, & \text{else.}\end{cases}$$

We utilize the following notation for central factorials

Definition 1.3 (Central factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0,\\ 1, & \text{if } k = 0,\\ n\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = n\left(n+\frac{k}{2}-1\right)_{k-1}, & \text{if } k > 0,\end{cases}$$

where $\left(n+\frac{k}{2}-1\right)_{k-1} = \prod_{j=0}^{k-2}\left(n+\frac{k}{2}-1-j\right)$ are falling factorials.

J. F. Steffensen gives quite a good discussion on factorial notations and definitions in [2].

2. Historical context

3. Auxiliary lemmas

Lemma 3.1 (Central Newton's formula)**.** *For a function f defined on integers and arbitrary integers x, t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes the k-th central finite difference

$$\delta^k f(t) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} f\left(t + \tfrac{k}{2} - j\right),$$

and $(x-t)^{[k]}$ are central factorials defined by (1.3).

Proof. See [3, p. 462], and [4, section 2, eq. (19)], and [5, section 6, eq. (21)]. □

Lemma 3.2 (Newton's formula for powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Lemma 3.3 (Central factorials binomial form)**.** *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k(k-1)!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

Lemma 3.4 (Central factorials binomial form decomposition)**.** *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \binom{n + \frac{k}{2}}{k} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k}\binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n+\frac{k}{2}}{k} = \frac{n+\frac{k}{2}}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

Thus,

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \binom{n+\frac{k}{2}}{k} - \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\left[2\binom{n+\frac{k}{2}}{k} - \binom{n+\frac{k}{2}-1}{k-1}\right] = \frac{1}{2}\left[\binom{n+\frac{k}{2}}{k} + \binom{n+\frac{k}{2}-1}{k} + \binom{n+\frac{k}{2}-1}{k-1} - \binom{n+\frac{k}{2}-1}{k-1}\right].$$

Thus, the statement follows. □

Lemma 3.5 (Newton's formula for powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}}{k} + \binom{n-t+\frac{k}{2}-1}{k}\right]\delta^k t^m.$$

Lemma 3.6 (Generalized hockey-stick identity)**.** *For integers a, b and j,*

$$\sum_{k=a}^{b}\binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^{b}\binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right)$. By hockey stick identity $\sum_{k=0}^{n}\binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^{b}\binom{k}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 3.7 (Centered hockey-stick identity)**.** *For integers $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $a = 1 - t + \frac{k}{2} + r$, and $b = n - t + \frac{k}{2} + r$ into Lemma (3.6) yields

$$\sum_{j=1}^{n} \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

□

4. Ordinary sums of powers

By decomposition of Newton's formula for powers (3.5), for integers $n, m \geq 0$, and an arbitrary integer t, we get

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^{m} \left[\sum_{j=1}^{n} \binom{j - t + \frac{k}{2}}{k} + \sum_{j=1}^{n} \binom{j - t + \frac{k}{2} - 1}{k} \right] \delta^k t^m. \tag{1}$$

Now we reach the transition to closed form of ordinary sums of powers. By setting $r = 0$, and $r = -1$ into Lemma (3.7), we get closed forms of the sums $\sum_{j=1}^{n} \binom{j-t+\frac{k}{2}}{k}$, and $\sum_{j=1}^{n} \binom{j-t+\frac{k}{2}-1}{k}$, respectively. Therefore,

Proposition 4.1 (Ordinary sums of powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^{m} \left[\binom{n - t + \frac{k}{2} + 1}{k+1} - \binom{1 - t + \frac{k}{2}}{k+1} + \binom{n - t + \frac{k}{2}}{k+1} - \binom{-t + \frac{k}{2}}{k+1} \right] \delta^k t^m.$$

For example,

Example 4.2 (Sums of squares in central differences)**.**

$$t = 0: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(1+n) + \frac{\delta^2 t^2}{2} \frac{n(1+3n+2n^2)}{6},$$

$$t = 1: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-1) + \frac{\delta^2 t^2}{2} \frac{n(1-3n+2n^2)}{6},$$

$$t = 2: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-3) + \frac{\delta^2 t^2}{2} \frac{n(13-9n+2n^2)}{6},$$

$$t = 3: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-5) + \frac{\delta^2 t^2}{2} \frac{n(37-15n+2n^2)}{6},$$

$$t = 4: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-7) + \frac{\delta^2 t^2}{2} \frac{n(73-21n+2n^2)}{6}.$$

Example 4.3 (Sums of cubes in central differences)**.**

$$t=0:\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(1+n) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(1+3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-1+n+4n^2+2n^3)}{24},$$

$$t=1:\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-1) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(1-3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(1+n-4n^2+2n^3)}{24},$$

$$t=2:\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-3) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(13-9n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-21+25n-12n^2+2n^3)}{24},$$

$$t=3:\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-5) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(37-15n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-115+73n-20n^2+2n^3)}{24},$$

$$t=4:\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-7) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(73-21n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-329+145n-28n^2+2n^3)}{24}.$$

It is indeed interesting to notice the connection between the decomposition of ordinary sums of powers (4.1) and Riordan's central factorial numbers of the second kind, that is

Proposition 4.4 (Central factorial numbers sums)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n+\frac{k}{2}+1}{k+1} - \binom{1+\frac{k}{2}}{k+1} + \binom{n+\frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1}\right] k!T(m,k),$$

where $T(n,k)$ denotes the central factorial numbers of the second kind (in Riordan's notation). See [5, p. 213]*, and* [6].

These central factorial numbers of the second kind $T(n,k)$ are defined by central Newton's formula for powers (3.2) evaluated in zero, such that

$$n^m = \sum_{k=0}^{\infty} T(m,k) n^{[k]}.$$

Which implies

$$T(n,k) = \frac{\delta^k 0^m}{k!},$$

by Newton's formula (3.2).

5. Remark on central factorial numbers of the second kind

In [1] Donald Knuth gives a remarkable formula for sums of odd powers

Proposition 5.1 (Knuth's odd powers sum)**.** *For non-negative integers n, m*

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k}{2k} T(2m, 2k),$$

where $T(2m, 2k)$ are central factorial numbers of the second kind.

It is quite interesting to notice that Proposition (5.1) originates from centered Newton's formula for powers (3.2) evaluated at zero. By Newton's formula, we have

$$n^{2m} = \sum_{k=0}^{2m} \frac{n^{[k]}}{k!} \cdot \delta^k 0^{2m}$$

The iteration for $k = 0$ can be omitted because $\delta^0 0^{2m} = 0$. Next, we observe that $\delta^k 0^{2m} = 0$ for every odd k, thus

$$n^{2m} = \sum_{k=1}^{m} \frac{n^{[2k]}}{(2k)!} \cdot \delta^{2k} 0^{2m}.$$

By definition of central factorial numbers of the second kind, yields

$$n^{2m} = \sum_{k=1}^{m} n^{[2k]} T(2m, 2k).$$

By applying identity (3.3), we get

$$\begin{aligned} n^{2m} = \sum_{k=1}^{m} \frac{(2k)!}{(2k)!} n^{[2k]} T(2m, 2k) &= \sum_{k=1}^{m} (2k)! \frac{n}{2k} \binom{n+k-1}{2k-1} T(2m, 2k) \\ &= \sum_{k=1}^{m} (2k-1)! n \binom{n+k-1}{2k-1} T(2m, 2k). \end{aligned}$$

By factoring out n, we obtain

$$n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k-1}{2k-1} T(2m, 2k).$$

By hockey stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ yields

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k}{2k} T(2m, 2k),$$

which is exactly the Proposition (5.1). Clearly, sum of odd powers above can be generalized to multifold case, by applying hockey stick pattern repeatedly

Proposition 5.2 (Knuth's odd powers sum multifold)**.** *For non-negative integers r, n, m*

$$\Sigma^{r+1} n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k+r}{2k+r} T(2m, 2k),$$

where $T(2m, 2k)$ are central factorial numbers of the second kind.

6. Generalized central factorial numbers of the second kind

Definition 6.1 (Generalized central factorial numbers of the second kind)**.** *For integers $0 \leq k \leq m$, and an arbitrary integer t, the generalized central factorial numbers of the second kind are defined by the relation*

$$n^m = \sum_{k=0}^{\infty} T_t(m, k) \cdot (n-t)^{[k]}.$$

It is also straightforward that

$$T_t(m, k) = \frac{\delta^k t^m}{k!},$$

by Newton's formula (3.2). For example,

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	0	1							
2	0	0	1						
3	0	$\frac{1}{4}$	0	1					
4	0	0	1	0	1				
5	0	$\frac{1}{16}$	0	$\frac{5}{2}$	0	1			
6	0	0	1	0	5	0	1		
7	0	$\frac{1}{64}$	0	$\frac{91}{16}$	0	$\frac{35}{4}$	0	1	
8	0	0	1	0	21	0	14	0	1

Table 1. Values of generalized central factorial numbers $T_0(n, k)$. See also the sequences A395862, and A370703 in the OEIS [7].

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	1	1							
2	1	2	1						
3	1	$\frac{13}{4}$	3	1					
4	1	5	7	4	1				
5	1	$\frac{121}{16}$	15	$\frac{25}{2}$	5	1			
6	1	$\frac{91}{8}$	31	35	20	6	1		
7	1	$\frac{1093}{64}$	63	$\frac{1491}{16}$	70	$\frac{119}{4}$	7	1	
8	1	$\frac{205}{8}$	127	$\frac{483}{2}$	231	126	42	8	1

Table 2. Values of generalized central factorial numbers $T_1(n,k)$. See also the sequences A395860, and A395861 in the OEIS [7].

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	2	1							
2	4	4	1						
3	8	$\frac{49}{4}$	6	1					
4	16	34	25	8	1				
5	32	$\frac{1441}{16}$	90	$\frac{85}{2}$	10	1			
6	64	$\frac{931}{4}$	301	190	65	12	1		
7	128	$\frac{37969}{64}$	966	$\frac{12411}{16}$	350	$\frac{371}{4}$	14	1	
8	256	$\frac{6001}{4}$	3025	3003	1701	588	126	16	1

Table 3. Values of generalized central factorial numbers $T_2(n,k)$. See also the sequences A394466, and A395314 in the OEIS [7].

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	3	1							
2	9	6	1						
3	27	$\frac{109}{4}$	9	1					
4	81	111	55	12	1				
5	243	$\frac{6841}{16}$	285	$\frac{185}{2}$	15	1			
6	729	$\frac{12753}{8}$	1351	585	140	18	1		
7	2187	$\frac{372709}{64}$	6069	$\frac{53011}{16}$	1050	$\frac{791}{4}$	21	1	
8	6561	$\frac{167943}{8}$	26335	$\frac{35049}{2}$	6951	1722	266	24	1

Table 4. Values of generalized central factorial numbers $T_3(n,k)$. See also the sequences A396559, and A395861 in the OEIS [7].

t	$T_t(2m,2k)$	OEIS ID
0	$1,0,1,0,1,1,0,1,5,1,0,1,21,14,1,0,1,85,147,30,1,\cdots$	A269945
1	$1,1,1,1,7,1,1,31,20,1,1,127,231,42,1,1,511,\cdots$	A394692
2	$1,4,1,16,25,1,64,301,65,1,256,3025,1701,126,1,\cdots$	A395456
3	$1,9,1,81,55,1,729,1351,140,1,6561,26335,6951,266,1,\cdots$	A395457

Table 5. Sequences of $T_t(2m,2k)$.

Thus, we have the formula for sums of powers in terms of generalized central factorial numbers of the second kind.

Proposition 6.2 (Generalized Central factorial sums)**.** *For integers $n,m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1}\right] k!T_t(m,k).$$

7. Double and triple sums of powers in terms of central differences

Proposition 7.1 (Double sums of powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^2 n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[+ \binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2}\Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+1}\Sigma^1 n^0$$
$$+ \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2}\Sigma^0 n^0 - \binom{0-t+\frac{k}{2}}{k+1}\Sigma^1 n^0\Bigg]\delta^k t^m$$

For example,

Example 7.2 (Double sum of squares in central differences)**.**

$$t=0: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\,n(1+n) + \tfrac{\delta t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(1+n)^2(2+n)}{12},$$

$$t=1: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\,n(1+n) + \tfrac{\delta t^2}{2}\,\tfrac{n(-1+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n^2(-1+n^2)}{12},$$

$$t=2: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\,n(1+n) + \tfrac{\delta t^2}{2}\,\tfrac{n(-4-3n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(10+5n-4n^2+n^3)}{12},$$

$$t=3: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\,n(1+n) + \tfrac{\delta t^2}{2}\,\tfrac{n(-7-6n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(32+23n-8n^2+n^3)}{12},$$

$$t=4: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\,n(1+n) + \tfrac{\delta t^2}{2}\,\tfrac{n(-10-9n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(66+53n-12n^2+n^3)}{12}.$$

Proposition 7.3 (Triple sums of powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^3 n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[$$
$$+ \binom{n-t+\frac{k}{2}+3}{k+3} - \binom{3-t+\frac{k}{2}}{k+3}\Sigma^0 n^0 - \binom{2-t+\frac{k}{2}}{k+2}\Sigma^1 n^0 - \binom{1-t+\frac{k}{2}}{k+1}\Sigma^2 n^0$$
$$+ \binom{n-t+\frac{k}{2}+2}{k+3} - \binom{2-t+\frac{k}{2}}{k+3}\Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+2}\Sigma^1 n^0 - \binom{0-t+\frac{k}{2}}{k+1}\Sigma^2 n^0\Bigg]\delta^k t^m$$

For example,

Example 7.4 (Triple sum of squares in central differences)**.**

$$t=0:\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(6+11n+6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(18+45n+40n^2+15n^3+2n^4)}{120},$$

$$t=1:\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-2-n+2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(-2-5n+5n^3+2n^4)}{120},$$

$$t=2:\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-10-13n-2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(58+65n-5n^3+2n^4)}{120},$$

$$t=3:\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-18-25n-6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(198+255n+40n^2-15n^3+2n^4)}{120},$$

$$t=4:\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-26-37n-10n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(418+565n+120n^2-25n^3+2n^4)}{120}.$$

8. Multifold sums of powers in terms of central differences

Proposition 8.1 (Multifold sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0\Bigg\}\delta^k t^m.$$

For example,

Example 8.2 (Examples of multifold sums)**.**

$$t=0: \quad \Sigma^r\, n^m = \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\tbinom{n+\frac{k}{2}+r}{k+r} + \tbinom{n+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\Big[\tbinom{r-s+\frac{k}{2}}{k+r-s} + \tbinom{r-1-s+\frac{k}{2}}{k+r-s}\Big]\Sigma^s\, n^0\Bigg\}\,\delta^k 0^m,$$

$$t=1: \quad \Sigma^r\, n^m = \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\tbinom{n-1+\frac{k}{2}+r}{k+r} + \tbinom{n-1+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\Big[\tbinom{r-s-1+\frac{k}{2}}{k+r-s} + \tbinom{r-2-s+\frac{k}{2}}{k+r-s}\Big]\Sigma^s\, n^0\Bigg\}\,\delta^k 1^m,$$

$$t=2: \quad \Sigma^r\, n^m = \tfrac{1}{2}\sum_{k=0}^{m}\Bigg\{\tbinom{n-2+\frac{k}{2}+r}{k+r} + \tbinom{n-2+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\Big[\tbinom{r-s-2+\frac{k}{2}}{k+r-s} + \tbinom{r-3-s+\frac{k}{2}}{k+r-s}\Big]\Sigma^s\, n^0\Bigg\}\,\delta^k 2^m.$$

9. Multifold sums of powers in generalized central factorial numbers

Proposition 9.1 (Multifold central factorial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r\, n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\Bigg[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\Bigg]\Sigma^s\, n^0\Bigg\} k!\,T_t(m,k).$$

Example 9.2 (Sums of squares in central factorial numbers)**.**

$$
\begin{aligned}
t=0: &\quad \Sigma^1 n^2 = n\cdot 0 + \tfrac{1}{2}n(1+n)\cdot 0 + \tfrac{1}{6}n(1+3n+2n^2)\cdot 1,\\
t=1: &\quad \Sigma^1 n^2 = n\cdot 1 + \tfrac{1}{2}n(n-1)\cdot 2 + \tfrac{1}{6}n(1-3n+2n^2)\cdot 1,\\
t=2: &\quad \Sigma^1 n^2 = n\cdot 4 + \tfrac{1}{2}n(n-3)\cdot 4 + \tfrac{1}{6}n(13-9n+2n^2)\cdot 1,\\
t=3: &\quad \Sigma^1 n^2 = n\cdot 9 + \tfrac{1}{2}n(n-5)\cdot 6 + \tfrac{1}{6}n(37-15n+2n^2)\cdot 1,\\
t=4: &\quad \Sigma^1 n^2 = n\cdot 16 + \tfrac{1}{2}n(n-7)\cdot 8 + \tfrac{1}{6}n(73-21n+2n^2)\cdot 1,\\
t=5: &\quad \Sigma^1 n^2 = n\cdot 25 + \tfrac{1}{2}n(n-9)\cdot 10 + \tfrac{1}{6}n(121-27n+2n^2)\cdot 1.
\end{aligned}
$$

From the example above, we may observe that

- For $t=0$ the coefficients $0,0,1$ correspond to the second row of the triangle (1).
- For $t=1$ the coefficients $1,2,1$ correspond to the second row of the triangle (2).
- For $t=2$ the coefficients $4,4,1$ correspond to the second row of the triangle (3).
- For $t=3$ the coefficients $9,6,1$ correspond to the second row of the triangle (4).

Example 9.3 (Sums of cubes in central factorial numbers)**.**

$$\begin{aligned}
t=0:\quad \Sigma^1 n^4 &= n\cdot 0 + \tfrac{1}{2}n(1+n)\cdot 0 + \tfrac{1}{6}n(1+3n+2n^2)\cdot 1\\
&\quad + \tfrac{1}{8}n(-1+n+4n^2+2n^3)\cdot 0 + \tfrac{1}{10}n(-2-5n+5n^3+2n^4)\cdot 1,\\
t=1:\quad \Sigma^1 n^4 &= n\cdot 1 + \tfrac{1}{2}n(n-1)\cdot 5 + \tfrac{1}{6}n(1-3n+2n^2)\cdot 7\\
&\quad + \tfrac{1}{8}n(1+n-4n^2+2n^3)\cdot 4 + \tfrac{1}{10}n(-2+5n-5n^3+2n^4)\cdot 1,\\
t=2:\quad \Sigma^1 n^4 &= n\cdot 16 + \tfrac{1}{2}n(n-3)\cdot 34 + \tfrac{1}{6}n(13-9n+2n^2)\cdot 25\\
&\quad + \tfrac{1}{8}n(-21+25n-12n^2+2n^3)\cdot 8\\
&\quad + \tfrac{1}{10}n(18-45n+40n^2-15n^3+2n^4)\cdot 1,\\
t=3:\quad \Sigma^1 n^4 &= n\cdot 81 + \tfrac{1}{2}n(n-5)\cdot 111 + \tfrac{1}{6}n(37-15n+2n^2)\cdot 55\\
&\quad + \tfrac{1}{8}n(-115+73n-20n^2+2n^3)\cdot 12\\
&\quad + \tfrac{1}{10}n(298-275n+120n^2-25n^3+2n^4)\cdot 1
\end{aligned}$$

From the example above, we may observe that

- For $t=0$ the coefficients $0,0,1,0,1$ correspond to the fourth row of the triangle (1).
- For $t=1$ the coefficients $1,5,7,4,1$ correspond to the fourth row of the triangle (2).
- For $t=2$ the coefficients $16,34,25,8,1$ correspond to the fourth row of the triangle (3).
- For $t=3$ the coefficients $81,111,55,12,1$ correspond to the fourth row of the triangle (4).

10. Multifold sums of powers in binomial form

Lemma 10.1 (Multifold sum of zero powers)**.** *For integers $r\ge 0$ and $n\ge 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

Proof. (1) Let $r=0$, then we have $\Sigma^0 n^0 = 1$, by definition (1.1).

(2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^{n} 1 = \binom{n}{1}$.

(3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^{n} \binom{k}{1} = \binom{n+1}{2}$.

(4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary integer r, by hockey stick identity $\sum_{k=1}^{n} \binom{k}{r} = \binom{n+1}{r+1}$, yields $\Sigma^r n^0 = \sum_{k=1}^{n} \Sigma^{r-1} k^0 = \sum_{k=1}^{n} \binom{k+r-2}{r-1} = \binom{r+n-1}{r}$.

□

Proposition 10.2 (Multifold binomial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \frac{1}{2} \sum_{k=0}^{m} \Bigg\{ \binom{n - t + \frac{k}{2} + r}{k + r} + \binom{n - t + \frac{k}{2} + r - 1}{k + r} - \sum_{s=0}^{r-1} \left[\binom{r - s - t + \frac{k}{2}}{k + r - s} + \binom{r - 1 - s - t + \frac{k}{2}}{k + r - s} \right] \binom{s + n - 1}{s} \Bigg\} \delta^k t^m$$

For example,

Example 10.3 (Examples of multifold binomial sums)**.**

$$t = 0: \quad \Sigma^r n^m = \tfrac{1}{2} \sum_{k=0}^{m} \Bigg\{ \binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1} \left[\binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \Bigg\} \delta^k 0^m,$$

$$t = 1: \quad \Sigma^r n^m = \tfrac{1}{2} \sum_{k=0}^{m} \Bigg\{ \binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1} \left[\binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \Bigg\} \delta^k 1^m,$$

$$t = 2: \quad \Sigma^r n^m = \tfrac{1}{2} \sum_{k=0}^{m} \Bigg\{ \binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1} \left[\binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \Bigg\} \delta^k 2^m.$$

11. Related works

12. Framework for sums of powers

In this manuscript, we derive formulas for sums of powers by combining centered Newton's interpolation formula for powers (3.2) with hockey-stick identities (3.6), and (3.7). This approach can be generalized further. In particular, while using Newton's formulas

$$\begin{cases} f(x) & = \sum_{k=0}^{\infty} \frac{(x-a)^{[k]}}{k!} \delta^k f(a) = \sum_{k=0}^{\infty} \frac{x-a}{k} \binom{x-a+\frac{k}{2}-1}{k-1} \delta^k f(a), \\ f(x) & = \sum_{k=0}^{\infty} \frac{(x-a)_k}{k!} \Delta^k f(a) = \sum_{k=0}^{\infty} \binom{x-a}{k} \Delta^k f(a), \\ f(x) & = \sum_{k=0}^{\infty} \frac{(x-a)^{(k)}}{k!} \nabla^k f(a) = \sum_{k=0}^{\infty} \binom{x-a+k-1}{k} \nabla^k f(a), \\ f(x) & = \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} \frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k} \end{cases} \tag{2}$$

one may replace linear difference operators δ, Δ, ∇ with their non-linear dynamic analogs, allowing more generic approach for sums of powers. It is indeed so remarkable that Taylor's series $f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} \frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k}$ is a continuous analogue of Newton's formula! Back to the topic, such non-linear difference operators arise naturally in the theory of dynamic equations on time scales, developed by Bohner and Peterson in [8]. Thus, the discrete change $x+h$ of a function $f(x+h)$ is replaced by function $g(h)$ which implies that dynamic difference operator is

$$D_g f = \frac{f(g(x)) - f(x)}{g(x) - x}. \tag{3}$$

A well-known example is Jackson's q-difference [9]

$$D_q f(x) = \frac{f(qx) - f(x)}{qx - x},$$

where $g(x) = qx$. In particular, for Newton's formula in linear differences δ, Δ, ∇, their respective falling, rising, and central factorials are related to binomial coefficients via

$$\begin{cases} \frac{(x)_n}{n!} & = \frac{1}{n!}x(x-1)(x-2)\cdots(x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} & = \frac{1}{n!}x(x+1)(x+2)\cdots(x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} & = \frac{1}{n!}\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = \frac{x}{n}\binom{x+\frac{n}{2}-1}{n-1}. \end{cases} \tag{4}$$

Motivated by this viewpoint, we can see that dynamic version of Newton's formula is

$$f(x) = \sum_{k=0}^{\infty} \frac{(x \mid \alpha, \beta)_k}{k!} \cdot D_g f(a), \tag{5}$$

where a is an arbitrary integer, $D_g f(a)$ is dynamic difference operator, and $(x \mid \alpha, \beta)_n$ is generalized factorial which uses the combination of Nörlund's notation [10], and definition given by Steffensen [2]

Proposition 12.1 (Generalized factorial)**.** *For non-negative integers x, n, α, β*

$$(x \mid \alpha, \beta)_n = x(x - n\alpha - \beta)(x - n\alpha - 2\beta)\cdots(x - n\alpha - n - I\beta).$$

For instance,

$$\begin{cases} (x)_n & = (x \mid \alpha, \beta)_n, \quad \alpha = 0, \quad \beta = 1, \quad \text{(forward Newton's formula)} \\ x^{(n)} & = (x \mid \alpha, \beta)_n, \quad \alpha = 0, \quad \beta = -1, \quad \text{(backward Newton's formula)} \\ x^{[n]} & = (x \mid \alpha, \beta)_n, \quad \alpha = \frac{1}{2}, \quad \beta = 1, \quad \text{(central Newton's formula)} \\ x^n & = (x \mid \alpha, \beta)_n, \quad \alpha = 0, \quad \beta = 0, \quad \text{(Taylor's formula)} \end{cases}$$

More formally, the algorithm derive closed form formula for sums of powers is:

Algorithm 12.2. *(1) Define the function $f(x) = x^m$.*

(2) Pick generalized difference operator $D_g f(x)$, see (3).

(3) Derive generalized Newton's formula (5) in terms of $D_g f(x)$, for example (2).

(4) Find an identity that connects respective factorials $(x \mid \alpha, \beta)_n$ with binomial coefficients (as per Newton's formula for powers), for example (4).

(5) Apply hockey stick identities (3.6), and (3.7) to derive closed form for sums of powers $\Sigma^1 n^m$.

(6) Apply identities (3.6), and (3.7) r-times to reach r-fold closed form for sums of powers $\Sigma^r n^m$.

13. Conclusions

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14. Mathematica programs

Use this `GitHub` repository to validate the results using Mathematica programs.

Mathematica Function	**Validates / Prints**
`MultifoldSumOfPowersRecurrence[r, n, m]`	Computes $\Sigma^r n^m$, (1.1)
`CentralFactorial[n, k]`	Computes $n^{[k]}$, (1.3)
`CentralFactorialsBinomialFormDecomposition[20]`	Validates Lem. (3.4)
`ValidateNewtonsFormulaForPowers[10]`	Validates Lem. (3.5)
`ValidateCenteredHockeyStickIdentity[5]`	Validates Lem. (3.7)
`ValidateOrdinarySumsOfPowersDecomposition[10]`	Validates Prop. (4.1)
`ValidateDoubleSumsOfPowersDecomposition[5]`	Validates Prop. (7.1)
`ValidateTripleSumsOfPowersDecomposition[5]`	Validates Prop. (7.3)
`ValidateMultifoldSumsOfPowers[3]`	Validates Prop. (8.1)